UW-IBM Hackathon Prompt Intro

IBM **Quantum**

Quantum Machine Learning Hackathon

"When does quantum data encoding ***actually help?"***

Using pre-computed data from IBM Heron R2 quantum hardware, investigate when, why, and how much quantum encoding helps – and do your own experiment with IBM quantum hardware!

Not a problem with a fixed answer — a mini research project where you draw your own evidence-based conclusions.

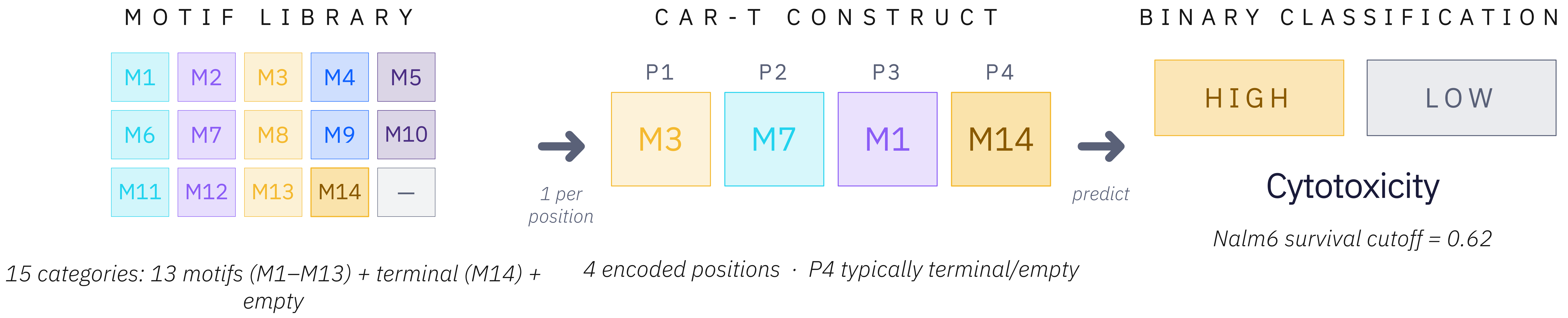
CAR T-Cell Immunotherapy Data

IBM Heron R2 60-qubit encoding

Python
Qiskit, Sklearn
IBM Quantum Account

CAR T-Cell Cytotoxicity Prediction

Binary classification of engineered immune cells from their molecular design — a sparse-data regime.



246 */ ~2,500*

Constructs measured out of all possible designs — ~10% sparse data regime

172 / 74 *split*

Train / test samples; binary task (Nalm6 survival < 0.62 = HIGH)

4 × 15 *= 60-dim*

Encoded positions × categories per position → one-hot input vector

Official guide tutorial : [Enhance feature classification using projected quantum kernel] Tutorial, IBM Quantum Cloud

References: Utro et al. *(2025)* *– main reference,* Daniels et al., *Science (2022)* *— experimental data*

Why Quantum Encoding Might Help Here

Three structural features of the CAR-T problem that **might** favor a quantum kernel — all hypotheses the experiments are designed to test.

01

Combinatorial Structure

Motifs may interact across positions.

The ZZ Feature Map entangles qubits — the encoding can depend on motif combinations across positions, not just individual identities. Track 2-C tests for this structure.

VS *Classical: one-hot encodes each motif × position independently; cross-position effects are left for the kernel to discover.*

02

Sparse-Data Regime

Only ~10% of the design space is measured.

246 of ~2,500 possible constructs have been tested. Theory frames sparse-data settings as where a different similarity structure could matter most — a hypothesis Track 2 puts to the test.

VS *Classical ML's strength is leveraging large N — exactly what's hard to obtain here.*

03

A Different Geometry

The kernel sees similarity differently.

60 qubits → X·Y·Z measurements → 180-dim projection. Constructs close in one-hot space may land far apart after projection — what Huang et al. (2021) quantify as the geometric difference g.

VS *Classical: RBF measures distance directly in one-hot space — no reshaping.*

Three Tracks, One Question

Required baseline + at least one experiment from Track 1 or Track 2.

TRACK 0 REQUIRED

Establish the Baseline

Train RBF-SVM on classical 60-dim features and on quantum-projected 180-dim features. Compare F1 scores and kernel heatmaps.

WHAT TO RECORD

Classical F1 · Quantum-projected F1 · ΔF1 · Kernel structure -> same as our tutorial

TRACK 1

Anatomy of the Quantum Encoding

What parts of the quantum encoding matter?

EXAMPLE EXPERIMENTS

- Which measurement basis (X, Y, Z) carries the signal?
- Where does quantum encoding help most across positions?
- Can fewer features preserve the benefit?

TRACK 2

Data Scarcity and Robustness

When does the quantum encoding help?

EXAMPLE EXPERIMENTS

- Learning curves: does scarcity widen the gap?
- Robustness: how much noise can it tolerate?
- Combinatorial structure: does motif arrangement matter?

What Students Will Gain

Open-ended exploration with real quantum hardware data — no “right” answers, only evidence-based discoveries.

Real Quantum Hardware Data

Work directly with 180-dim projections pre-computed on IBM Heron R2 from a 60-qubit ZZ Feature Map circuit. Not simulations — actual QPU outputs, ready for sklearn analysis.

Low Floor, High Ceiling

All you need: Python, pandas, scikit-learn, matplotlib. No quantum mechanics background required — focus on data science and experimental thinking.

Time to be a Scientist

Follow the Expectation → Validation → Limitation → Attempt framework. The goal: one evidence-based claim about whether, when, and why quantum encoding helps.

A Problem That Matters

CAR T-cell signaling motif design for cancer immunotherapy — a domain where data is expensive and sparse. The genuine open question: when does quantum encoding actually help, and when doesn’t it?

— Baseline + one well-designed experiment + one evidence-based claim — that’s a good hackathon outcome.

Judging Criteria (draft)

Track 0 Completion (25 pts)

- Both pipelines working
- F1 scores reported
- Kernel visualizations

Originality (30 pts)

- Creative experiments
- Novel approaches
- Insights beyond requirements

Presentation (25 pts)

- Clear structure
- Good visuals
- Time management

Scientific Rigor (20 pts)

- Proper experimental design
- Statistical validity
- Honest reporting

Bonus (+20 pts)

- Geometric difference (+5)
- QPU experiments (+10)
- Novel insights (+5)

Total: 100 pts + 20 bonus